Question 1

Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

Answer

1. **Prepare.** Save your work and plug the webcam directly into a USB port on the computer (not through a hub). Remove any lens cap or privacy cover.
2. **Connect.** Insert the USB cable. Windows shows a *Setting up a device* notification and installs a standard (UVC) driver automatically.
3. **Check the driver.** Press Win + X and open *Device Manager*. Expand *Cameras* or *Imaging devices* and confirm the webcam is listed with no yellow warning icon.
4. **Install the manufacturer driver if needed.** If it is not listed or lacks features, download the driver for your exact webcam model and Windows version from the maker's website and run the installer (or use Settings > Update & Security > Windows Update > *Check for updates*). Restart the PC if asked.
5. **Allow camera access.** Open Settings > Privacy > Camera and make sure *Allow apps to access your camera* is ON.
6. **Open the Camera app.** Click Start, type *Camera* and press Enter. If the PC also has a built-in camera, click *Change camera* to select the external webcam. A live video preview should appear.
7. **Capture a photo.** Make sure *Photo* mode is selected and click the shutter button.
8. **Verify.** Click the thumbnail (or open Pictures > Camera Roll) and check the picture is clear and saved. This confirms the webcam and driver work.

Check	Expected result
Device Manager	Webcam listed under Cameras / Imaging devices, no warning icon
Camera app	Live preview visible
Photo captured	Image saved in Pictures > Camera Roll

Question 2

Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Answer

1. **Prepare the printer.** Unpack it, remove the packing tape, insert ink or toner, load paper and plug in the power cable.
2. **Connect it.** *USB:* connect the USB cable (Type-B end to the printer, Type-A end to the PC) and switch the printer on. *Network:* join the printer to Wi-Fi or Ethernet from its control panel and note its IP address (print a network configuration page).
3. **Add the printer in Windows.** Open Settings > Devices > Printers & scanners > *Add a printer or scanner*. Wait for the printer to appear, then click *Add device*.
4. **For a network printer that is not listed:** click *The printer that I want isn't listed* > *Add a printer using a TCP/IP address or hostname* > type the IP address > Next.
5. **Install the correct driver.** Let Windows install it, or download the driver for the exact printer model and Windows version (32-bit or 64-bit) from the manufacturer's website and run the installer. In the wizard you can also choose the manufacturer and model manually, or use *Have Disk*.
6. **Set as default (optional).** Select the printer and click *Manage* > *Set as default*.
7. **Print a test page.** Click the printer > *Manage* > *Print a test page* (or Printer properties > *Print Test Page*).
8. **Confirm success.** The page prints showing the printer name and driver details, Windows asks whether it printed correctly (choose *Close*), and the status shows *Ready*.

Connection type	Cable / method	Key setup detail
USB	USB Type-A to Type-B cable	Plug in and Windows detects it automatically
Network	Wi-Fi or Ethernet	Add by IP address if it is not found automatically

Question 3

Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

Answer

1. **Start from a working setup.** Confirm the printer shows *Ready* in Printers & scanners and appears in Device Manager under *Print queues* or *Printers*.
2. **Simulate the fault.** Unplug the USB cable from the printer (or the PC). Windows plays a disconnect sound, the printer turns grey or shows *Offline*, and a print job gets stuck.
3. **Open Device Manager.** Press Win + X and choose *Device Manager*.
4. **Identify the problem.** Look for an error icon (yellow triangle with an exclamation mark), an *Unknown device* or *Other devices* entry, an *Unknown USB Device* under Universal Serial Bus controllers, or the printer missing altogether.
5. **Resolve the connection.** Plug the cable back in firmly. If needed, use a different USB port or cable (avoid hubs) and make sure the printer is switched on.
6. **Scan for hardware changes.** Right-click the computer name at the top of Device Manager (or open the *Action* menu) and choose *Scan for hardware changes*. The printer should reappear.
7. **If an error icon remains.** Right-click the device > *Properties* to read the error code, then choose *Update driver* > *Search automatically*. If that fails, choose *Uninstall device*, reconnect the cable and scan again, then restart the PC.
8. **Verify.** Check the printer shows *Ready* in Printers & scanners, and print a test page.

What you see in Device Manager	Meaning	Fix
Yellow triangle with !	Driver or hardware problem	Update or reinstall the driver
Unknown device / Unknown USB Device	Windows cannot identify the device	Change cable or port, scan for hardware changes
Printer not listed	Not connected or not powered	Reconnect, power on, then scan for hardware changes

Question 4

A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

Answer

Because the camera app works, the webcam hardware and driver are fine. The fault is in Zoom, a Windows permission, or another app holding the camera.

1. **Close apps that are using the webcam.** Only one app can use the webcam at a time. Close the Camera app, Teams, Skype, browser tabs (such as Google Meet) and OBS. Check Task Manager and end any that remain.
2. **Restart Zoom completely.** Right-click the Zoom icon in the system tray, choose *Quit*, then reopen it.
3. **Enable permissions.** Settings > Privacy > Camera: turn ON *Allow apps to access your camera* and *Allow desktop apps to access your camera*. Zoom is a desktop app, so the second switch is the one that matters.
4. **Select the right camera in Zoom.** Open Zoom Settings > Video and choose the webcam from the *Camera* drop-down (not a virtual camera such as OBS). During a meeting, click the arrow beside *Start Video* and check *Stop Video* is not active.
5. **Update.** Update Zoom (profile picture > *Check for updates*) and update the webcam driver in Device Manager. Reinstall Zoom if the problem continues.
6. **Restart the PC** to clear any process still locking the camera.
7. **Verify in both apps.** Open Zoom Settings > Video and confirm the live preview appears. Then close Zoom and open the Camera app to confirm it still works. Use them one at a time, as the webcam cannot be shared by both at once.

Possible cause	How to check	Fix
Another app using the webcam	Task Manager / open apps	Close that app
Zoom blocked by privacy settings	Settings > Privacy > Camera	Turn on desktop app access
Wrong camera selected in Zoom	Zoom Settings > Video	Choose the correct webcam